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4. Memory Management

Imp

	paging	Segmentation
1)	program is divided into Fixed size pages.	program is divided into variable size segments.
2)	paging is faster than segmentation.	segmentation is slower than paging.
3)	paging is invisible to the user.	segmentation is visible to the user.
4)	processor use Segment number, offset to calculate absolute address	process use page number, offset to calculate absolute address.
5)	Logical address is positive integer.	Logical address is ordered pair (segment, offset).
6)	A page is of physical unit.	A page is of logical unit.
7)	Each process is assigned its page table.	Each process is assigned its segement table.
8)	No shared code	share code.
9)	No separate protection	Seperate protection.
10)	seperate compiling.	seperate compiling.

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Study material provided by: Vishwajeet Londhe

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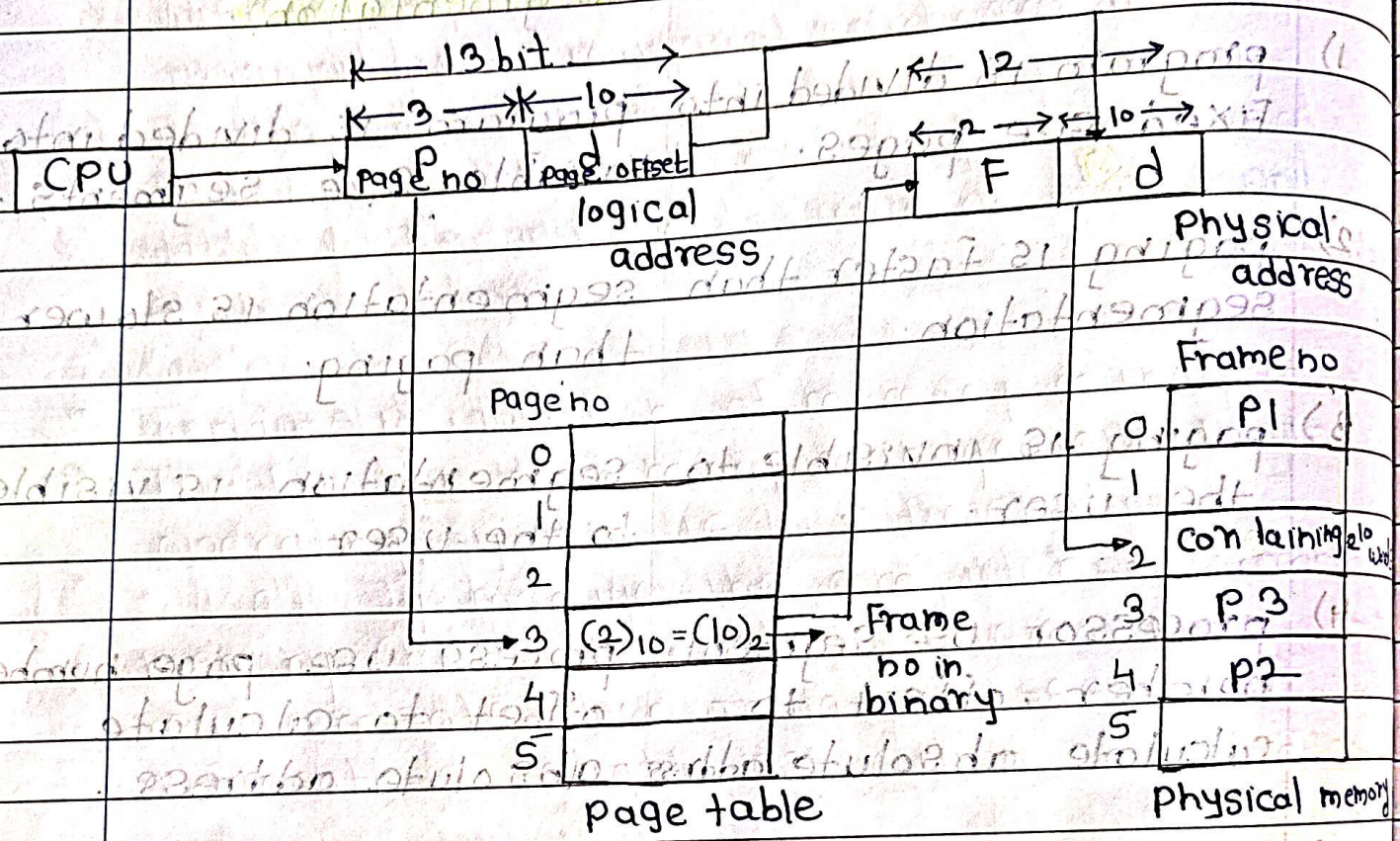
Paging - paging is a storage mechanism used as to retrieve processes from secondary storage to main memory as pages.



- paging is a memory management scheme.
- paging reduces the external fragmentation.
- Memory is divided into fixed size blocks called page frames.
- The virtual address space of process is also spilt into fixed size blocks of the same size called pages.
- The size of the page frame is determined by the Hardware.
- when a process is loaded into memory, the operating system loads each page into an unused page frame.
- CPU generates the logical address & it consists

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of two parts : page number (P) & page offset(d)
- In case of simple partition, a logical address is the location of a word relative to the beginning of the program, the processor that into a physical address.
- page table store the number of the page frame allocated for each page.
- The page number is used as index into a page table.
- page table contains the base address is combined with the page offset to define the physical memory address that is sent to the memory unit.
- The size of page is typically a power of 2, varying between 512 bytes & 16MB per page depending on the computer architecture.
- page size is always a power of 2 because all addresses are binary and are divided into page or page frame number & offset.
- The physical address can be generated by the offset of the frame number logical address consists of page number & page offset.

page number (P) = m - n.
page offset (d) = n.
P = Index into page table.
d = offset within the page.

- Advantages of paging
- Support higher degree of multi-processing

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paging increase memory & processor utilization

Disadvantages

- page address mapping hardware usually increase the cost of the computer.
- Memory must be used to store the various tables like page table, memory map table etc.

Imp

Explain the function of memory management

→ The function of memory management in an operating system is to efficiently & effectively manage the computer's primary memory (RAM) to meet the memory requirements of running processes & provide a smooth & reliable execution environment.

1) **Memory Allocation:**

- Memory management allocates memory to processes when they are created or when they request additional memory.
- It tracks availability & usage of memory blocks & assign suitable memory regions to processes.
- Different memory allocation strategies, paging or segmentation may be used depending on system design & requirements.

2) **Memory Deallocation:**

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- when a process terminates or no longer requires a specific memory region, memory management deallocation the memory & makes it available for other processes.
- It ensure the memory is properly released & marked as available for reuse.
- Deallocation may involve updating data structures such as page tables or free memory lists, to reflect the memory.

3) Memory protection -

Memory management enforce memory protection mechanisms to prevent unauthorized access or modification of memory.

- Each process is typically assigned its own protected memory space, isolating it from other processes.
- Memory process protection help ensure the stability & security of the system by preventing one process from accessing or modifying memory that belongs to another process.

4) Memory Mapping & Swapping -

- Memory management facilities memory mapping, allow processes to access files or devices as if they were part of the processes memory space.
- It manages the swapping of memory

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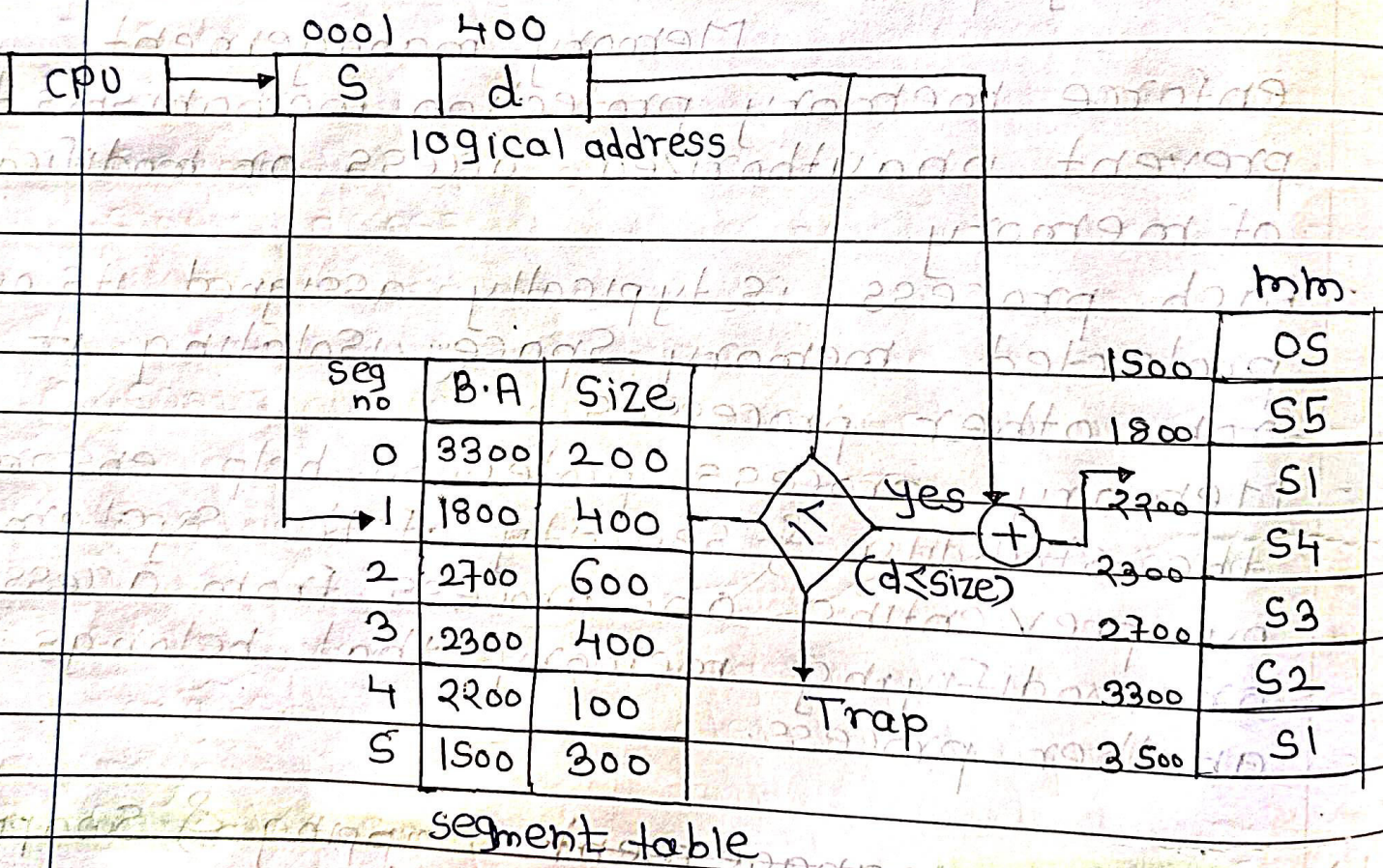
Space between primory memory (RAM) & secondary storage (such as hard disk) to support virtual memory systems.
- Swapping help overcome physical mem limitations by transparently moving less frequently used pages to Secondary storage & bringing back when needed.

10) Segementation :-



- Segementation divides the virtual address space of process into variable-size logical segements, such as code segment, data segment, stack segments etc.

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DOMS | Page No.
Date / /

- Each Segment is assigned a unique identifier, & the physical memory is divided into corresponding variable-sized segments or pages.
- The logical address is generated by a process consists of a segment identifier & an offset with the segment.
- To translate a logical address to a physical address:
- The segment identifier is used to index a Segment table, which store the base address & length of each segment.
- If the segment is larger than a single page, an additional page table is used to transalte the linear address to the physical address.
- Segementation is a memory management technique where a process is divided into variable size called "Segments".
- Variable-size divisions - segments can have different lengths, depending on the program's requirements.
- Logical division of memory - segments represent meaningful units like code, stack, data or modules.
- Two-part address - A logical address consist of
- segment number (S) - identifies which segment is being referred to.

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- offset (d) - specifies the exact location which segment.
logical address = (segment number, offset)

Advantages
- Smaller segment Table.
- Better CPU utilization
- security & seperation.
- closer to user's view.

Disadvantages
- overhead.
- complexity
- segmentation faults.
- slower access.
- External Fragmentation

Imp	Contiguous allocation	Non-contiguous allocation
1)	wastage of memory	No wastage of memory.
2)	suffer from internal Fragmentation	only paging, suffer from internal fragmentation.
3)	Allocates single area of memory for process.	Allocates more than one block of memory for process.
4)	program execution take place without overhead.	Address translation is overhead.

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5)	swapped-in processes are placed in the original area.	Swapped in processes can be placed any where in memory.

Imp

	Internal Fragmentation	External Fragmentation.
1)	Memory allocated to a process may be slightly larger than the requested memory. The difference between these two numbers is internal Fragmentation.	External Fragmentation exists when there is enough total memory space to satisfy a request, but available spaces are not contiguous.
2)	First fit & best fit memory allocation does not suffer from internal fragmentation.	First fit & best fit memory allocation suffers from external fragmentation.
3)	Systems with fixed sized allocation	The solution to external fragmentation is compaction & paging.
4)	paging's suffer from internal fragmentation	paging does not suffer from external fragmentation.

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Imp

What is internal fragmentation & external fragmentation :

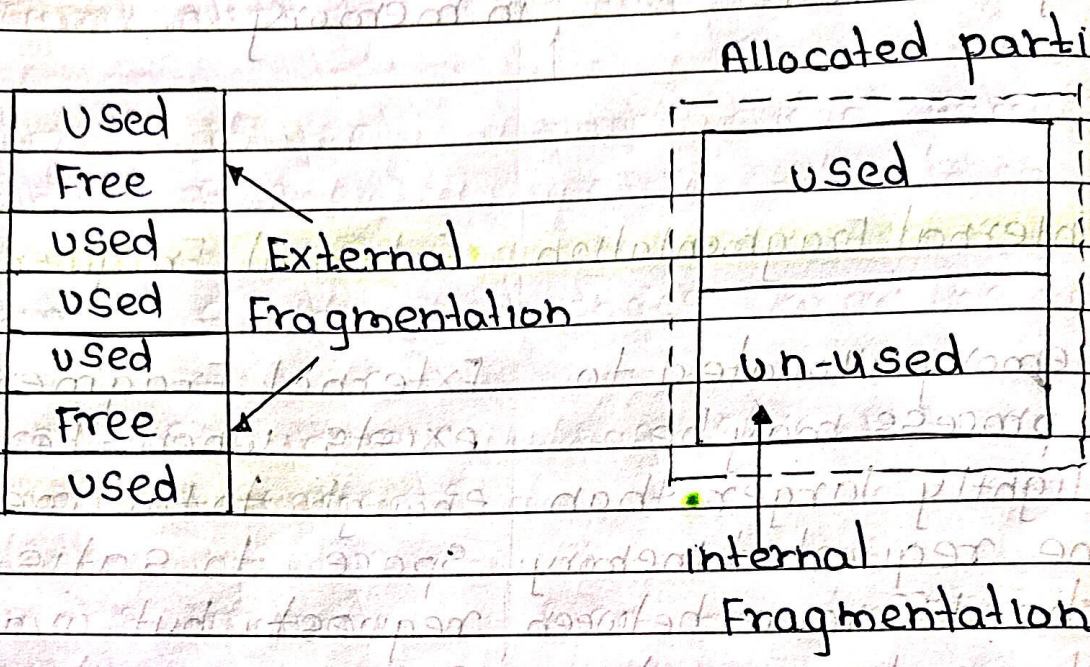

Fig. External & Internal Fragmentation.

- As processes are loaded and removed from memory, the free memory space is broken into little pieces.
- **Internal Fragmentation -**

Internal Fragmentation occurs when the memory allocator leaves extra space empty inside of a block of memory that has been allocated for a client.

- Internal fragmentation occurs when there is unused space within a memory block.
- e.g. if a system allocates a 64 kB blocks of memory to store a file that is only 400 kB in size, that block will contain

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2.4KB of internal fragmentation.
- When the system employs a fixed-size block allocation method, such as memory allocator with a fixed block size, this can occur.

- **External Fragmentation:-**
External fragmentation occurs when a strage medium, such as hard disc or Solid state drive, has many small blocks of free space scatter throughout it.
- This can happen when a system creates & deletes files frequently, leaving many small blocks of free space on the medium.
- When a system needs to store a new file, it may be unable to find a single contiguous block of free space large enough to store the file & much instead store the file in multiple smaller blocks.
- This can causes external fragmentation & performance problems when accessing the files.
- Fragmentation can also occur at various level within a system.
- Memory fragmentation can occur at the memory management level, where the system allocates & deallocated memory blocks dynamically.

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Imp

Comparsion of Demand paging with segmentation

Segmentation	Demand paging
1) Segments may of different size.	pages are of same size.
2) segment can be shared	pages cannot be shared.
3) It allows dynamic growth of segments.	page size is fixed.
4) provides virtual memory.	Also provide virtual memory.
5) Segments are allocated to the program while compilation.	pages are loaded in memory on demand.
6) Segment map table indicates the address of each segment in memory.	page map table keeps track of pages in memory.

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Imp.

• Define Virtual memory & how it is implemented?

• Virtual Memory :-

In many computer system, programmers often realize that some of their large programs cannot fit in main memory for execution.

- The virtual memory allows execution of partially loaded process.
- The ability to execute a partially loaded process is also advantages from the operating system points of views.

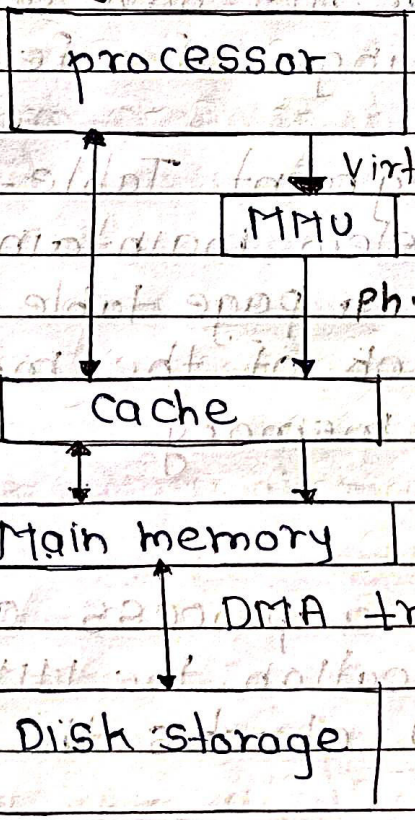

Fig. Virtual memory organization

- Virtual memory makes the task of programming much easier.
- Virtual memory is commonly implemented

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by demand paging.
- Virtual memory mechanism bridges the size & speed gaps between the main memory & Secondary Storage & is usually implemented
- Virtual memory is memory management technique used by operating systems to provide an illusion of having more physical memory (RAM) than is actually available.

1) paging or Segmentation -
- The virtual address space of process is divided into pages or segments.
- Each page or segments have a fixed size, typically in the range of a few kilobytes.

2) page Table or Segment Table -
- The operating system maintains a data structure called a page table or Segment table to keep track of the mapping between virtual & physical memory.

3) page Faults -
When a process accesses a virtual memory location, the MMU (memory management unit) checks the page table or segment table.
- This process is called page swapping or demand paging.

4) memory Mapping -

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The operating system use memory mapping techniques to associate files or devices with portions of the virtual address space.
- This allows processes to access files or devices as if they were regular memory locations, simplifying data transfer & interprocess communication.

IMP. What is Trashing? Cause of trashing?

→ Trashing refers to a situation in computer systems where excessive paging or swapping of pages between physical memory (RAM) & secondary storage (such as a hard disk) occurs, leading to a severe degration in system performance.
- It is characterized by a high rate of page faults & low effective CPU utilization.
- The primary causes of trashing is the excessive demand for memory resources by the running processes compared to the available physical memory capacity.

1) Insufficient physical memory -
- If the system does not have enough physical memory to accommodate the memory requirements of all active processes, trashing is more likely to occors.
- Insufficient physical memory forces the operating system to rely heavily on paging resulting in frequent page faults & increased disk I/o.

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2) overallocation of Memory
- If the system over commits or overallocates memory resources to multiple processes, each processes may receives an allocation larger than its actual memory needs.
- This can lead to excessive paging as the system tries to fulfill the inflated memory demands, resulting in trashing.

3) Memory Thrashing by processes
- In some cases the behavior of individual processes can lead to trashing.
- If a processes exhibits poor locality of reference, constantly accessing a large number of pages that does not reside in physical memory, it can trigger excessive paging and Trashing.

To mitigate trashing, several strategies can be employed such as,
- adjusting process
- Improving page replacement algorithm to better handle trashing situations.
- optimizing memory allocation & utilization
- Increasing the amount of physical memory available memory to the system

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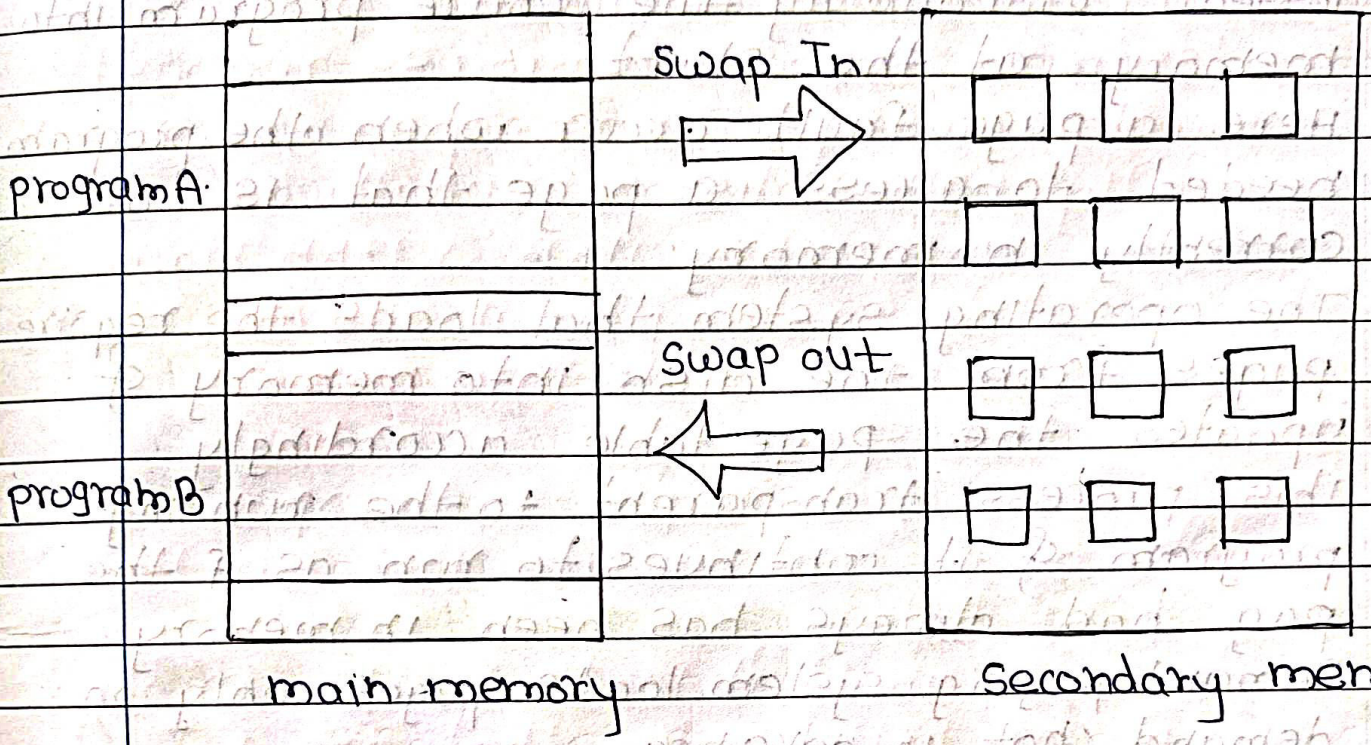
(Imp)

Demand paging:



- A demand paging is similar to a paging system with swapping.
- Demand paging combines the features of simple paging & overlaying to implement virtual memory.
- Demand paging is a virtual memory management technique in an operating systems where pages of a program are loaded into main memory only when they are needed, rather than all at once at the start.
- Demand paging is a technique used in virtual memory systems where pages enter main memory only when requested or needed

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by the CPU.
- OS loads only the necessary pages of a program into memory at runtime, instead of loading the entire program into memory at the start.
- Here, a page fault occur when the program needed to access a page that is not currently in memory.
- The operating system that loads the require pages from the disk into memory & updates the page table accordingly.
- This process transparent to the running program & it continues to run as if the pag had always has been in memory.
- Demand paging system loads pages only on demand 'not in advance.
- Instead of swapping the entire process into memory, the pages or the lazy swapper is used.
- How demand paging works
- Initial load - when a process starts, no pages are initially loaded into physical memory.
- Instruction execution - The CPU begains executing the program.
- The first instruction to access may be on a page that is not in memory.

page fault - The system memory management (MMU) detects that the required page is

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missing, triggering a page fault.

4) page retrival - The operating system takes control & locates the required pag on the disk.

5) Frame allocation - The os finds a free frame in the main memory.

6) page loading - The os load the requested page from the disk into the newly available frame.

IMP

Describe the following memory allocation (placement algorithm) strategies

i) First fit
2) Best fit
3) Worst fit

1) First fit - They may be many holes in the memory, so the operating system to reduce the amount of time it spends analyzing the available spaces, begins at the start of memory & allocates memory from the hole it encounters large enough to satisfy the request.

- Using the same example as above, first fit will allocate 12KB of the 14 KB block to the process.

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2) **Best fit**-

The allocators places a process in the smallest block of unallocated memory in which it will fit.

- For example, suppose a process requests 12KB of memory & memory manager currently has a list of unallocated blocks of 6KB, 14KB, 19KB, 11KB & 13KB blocks.
- The best fit strategy will allocate 12KB of the 13KB block of the process.

problem of best fit-

It requires an expensive search of the entire free list to find the best hole.

- More importantly, it leads to the creation of lots of little holes that are not big enough to satisfy any requests.
- This situation is called fragmentation & is a problem for all memory management strategies, although it is particularly bad for benefit.

Solution- one way to avoid making little holes is to give the client a bigger block than it asked.

- ex - we might round all requests upto the next larger multiple of 64 bytes.
- That does not make the fragmentation go away it just hides it.
- Unusable space in the form of holes is called external fragmentation.

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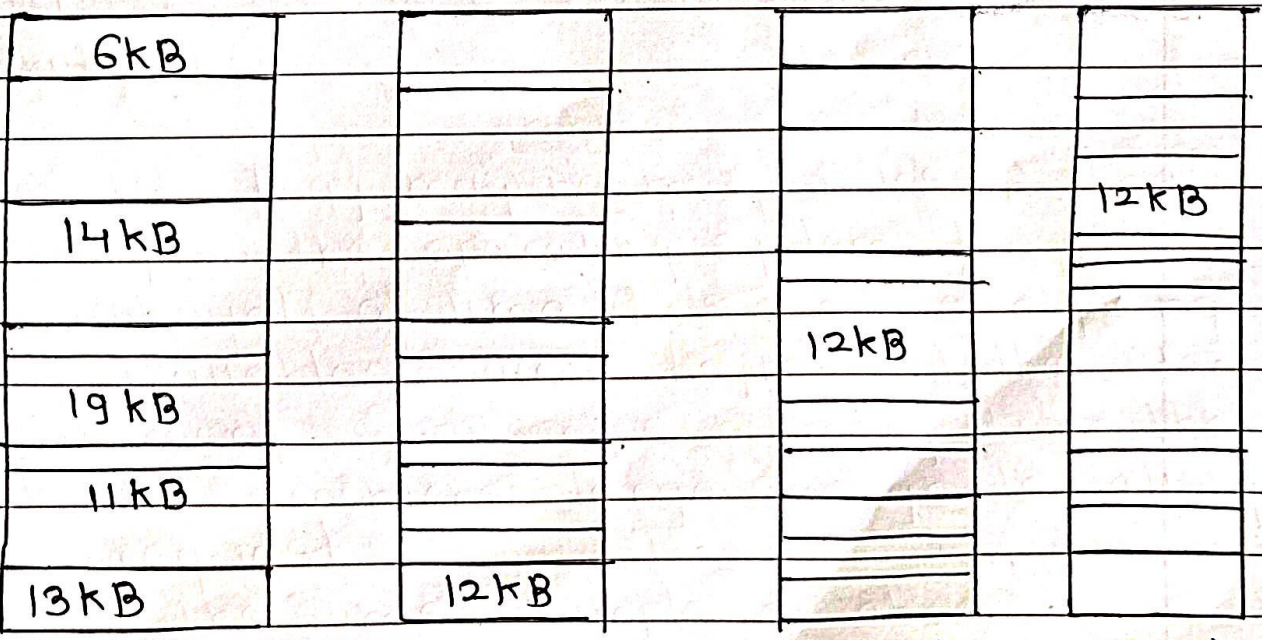
go away it just hides it.

- **Worst fit** - The memory manager places a process in the largest block of unallocated memory available.
- The idea is that is placement will create the largest hold after the allocation, thus increasing the possibility that compared to best fit, another process can use the remaining space.
- Using the same example as above, worst fit will allocate 12kB of the 19kB block to the process, leaving a 7kB block of future use.



placement algorithm.

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Assignment No-4

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1) what is the difference between paging & segementation.
2) Describe the following memory allocation strategies
   1) first fit
   2) Best fit
   3) worst fit
3) Explain the concept of paging with neat diagram?
4) Differentiate the contiguous & non-contiguous memory allocation.
5) Draw & Explain schematic of virtual memory?
6) Explain segmentation with suitable diagram?
7) Eloborate the concept of demand paging with appropriate diagram.
8) Explain Internal & External fragmentation?
9) Explain function of memory management?
10) what is Trashing ? causes of trashing?
11) Differentiate between segmentation & Demand paging?